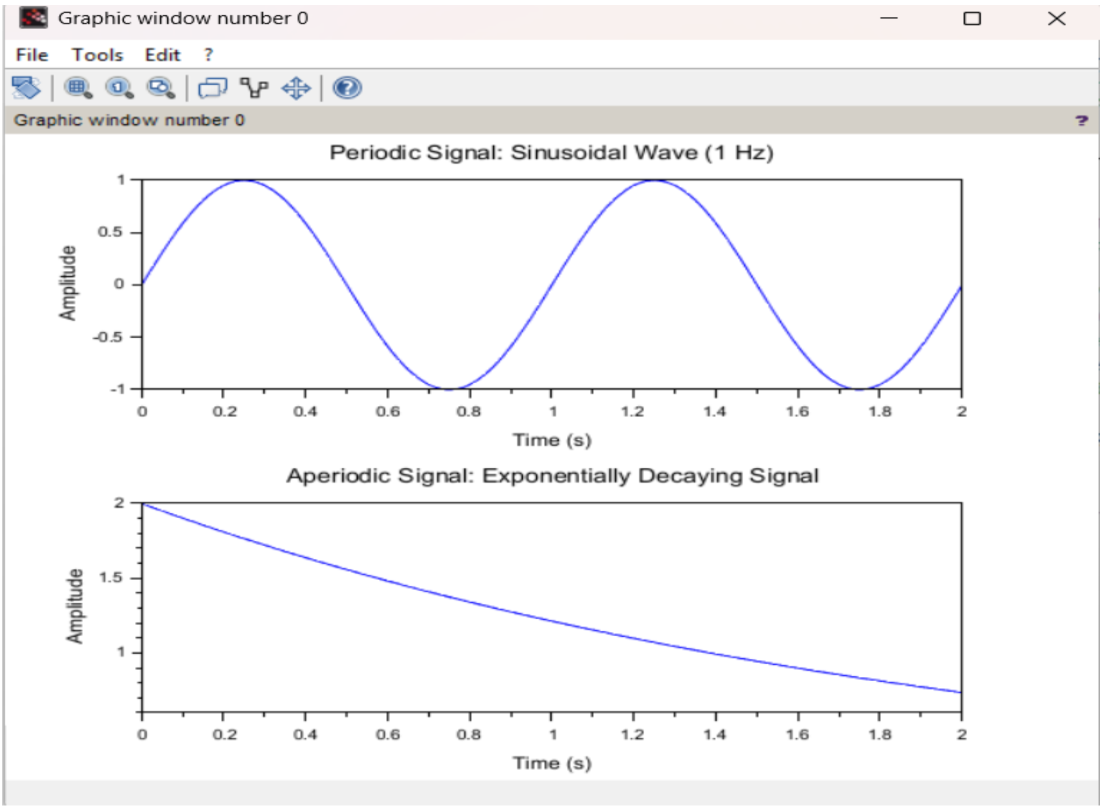
2. SIGNAL PROPERTIES AND TRANSFORMATIONS

Experiment 2.1

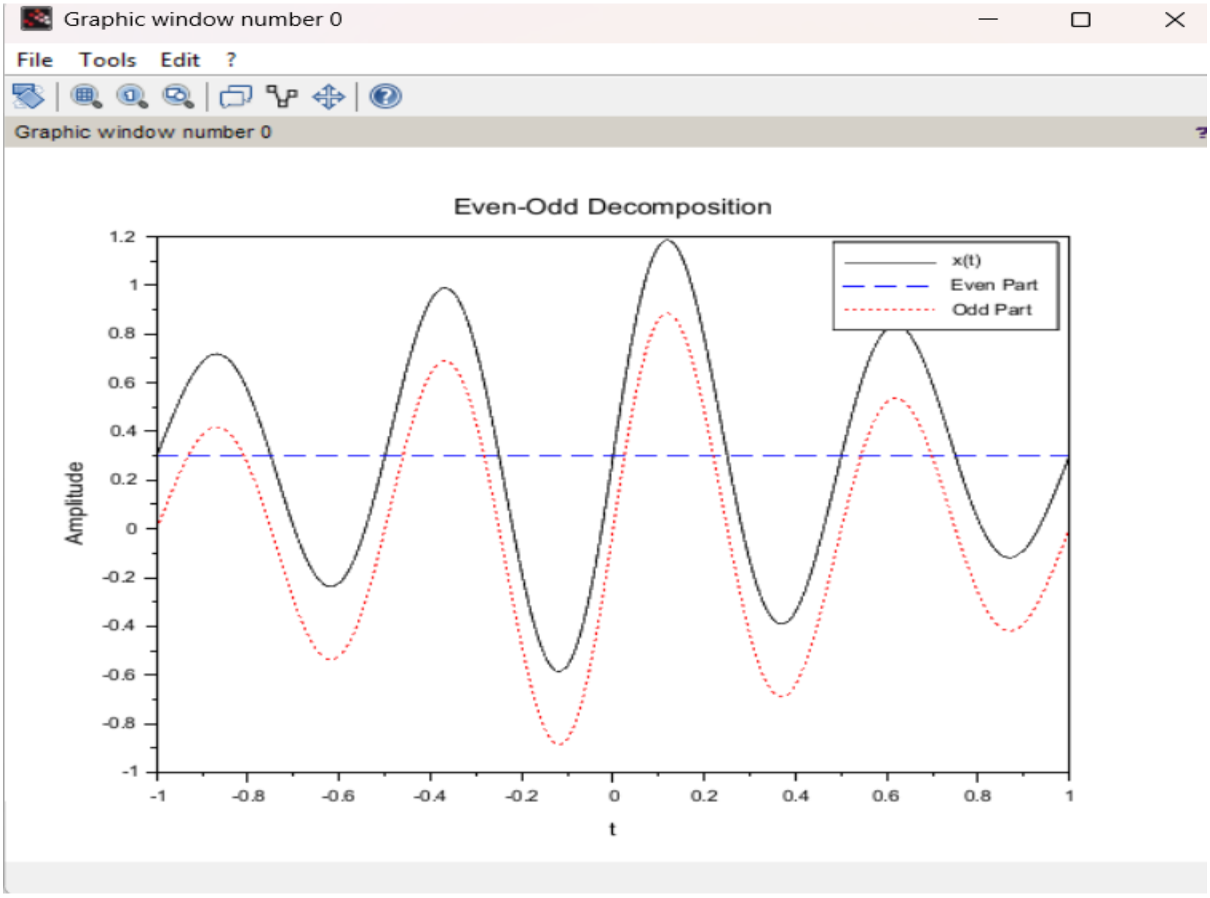
```
// Test Case: Generation of Periodic and Aperiodic Signals

// Parameters for the periodic signal (Sinusoidal)

f_periodic = 1; // Frequency in Hz

A_periodic = 1; // Amplitude

t_periodic = 0:0.01:2; // Time vector for 0 to 2 seconds

// Generate the periodic signal

x_periodic = A_periodic * sin(2 * %pi * f_periodic * t_periodic);

// Plot the periodic signal

clf; // Clear the current figure

subplot(2,1,1); // Subplot 1 for the periodic signal

plot(t_periodic, x_periodic);

xlabel('Time (s)');

ylabel('Amplitude');

title('Periodic Signal: Sinusoidal Wave (1 Hz)');

// Parameters for the aperiodic signal (Exponentially Decaying)

a_aperiodic = 0.5; // Base of the exponential decay
```

```
A_aperiodic = 2; // Amplitude
t_aperiodic = 0:0.01:2; // Time vector for 0 to 2
seconds
// Generate the aperiodic signal
x_aperiodic = A_aperiodic * exp(-a_aperiodic *
t_aperiodic);
// Plot the aperiodic signal
subplot(2,1,2); // Subplot 2 for the aperiodic signal
plot(t_aperiodic, x_aperiodic);
xlabel('Time (s)');
ylabel('Amplitude');
title('Aperiodic Signal: Exponentially Decaying Signal');
```



Experiment 2.2

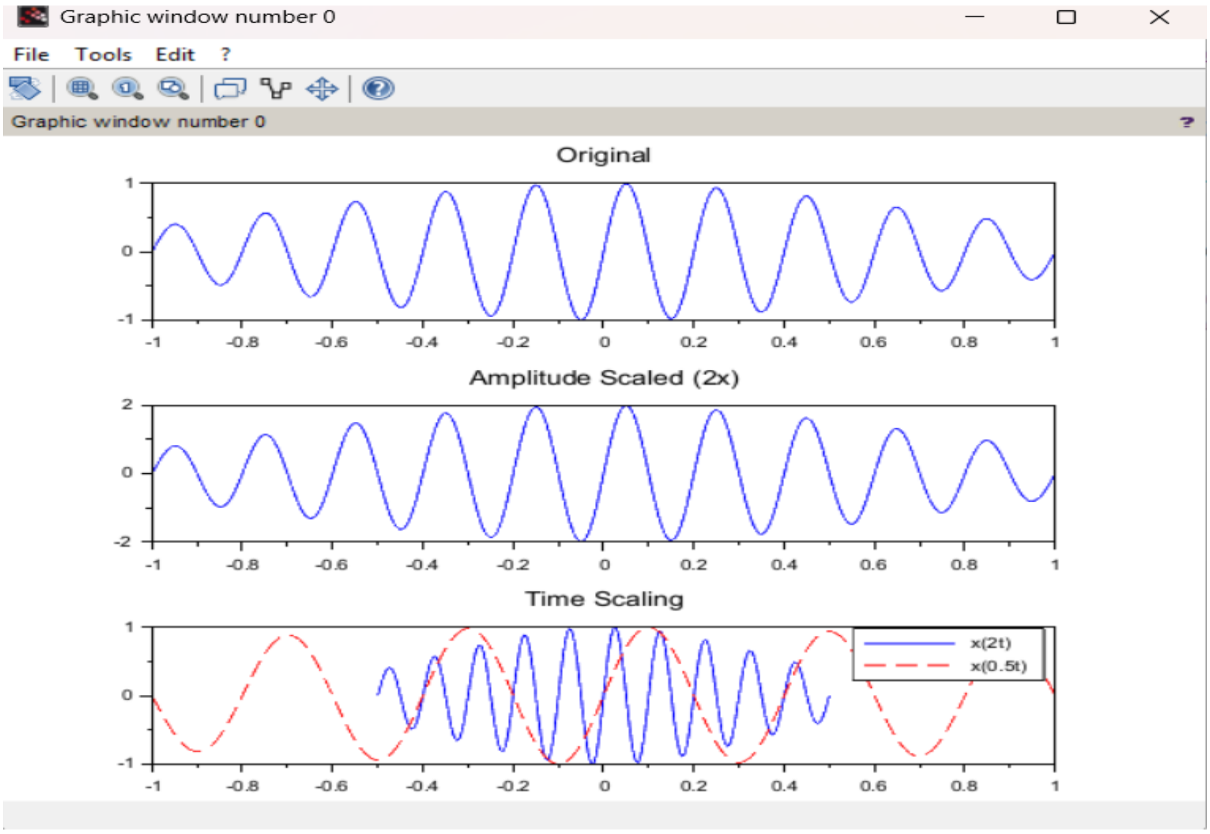
```
t = -1:0.001:1;
x = exp(-abs(t)).*sin(4*%pi*t) + 0.3;
// Reverse signal
x_rev = interp1(t, x, -t, 'linear');
// Even and odd parts
xe = 0.5 * (x + x_rev);
xo = 0.5 * (x - x_rev);
clf;
plot(t, x, 'k');
plot(t, xe, 'b--');
plot(t, xo, 'r:');
legend("x(t)", "Even Part", "Odd Part");
xtitle("Even-Odd Decomposition", "t", "Amplitude");
```



Experiment 2.3

```
t = -1:0.001:1;

x = sin(2*%pi*5*t).*exp(-t.^2);

// Amplitude scaling

x_amp = 2*x;

// Time scaling

x_time_comp = interp1(t, x, 2*t, 'linear');

x_time_exp = interp1(t, x, 0.5*t, 'linear');

clf;

subplot(3,1,1);
```

```
plot(t, x); xtitle("Original");
subplot(3,1,2);
plot(t, x_amp); xtitle("Amplitude Scaled (2x)");
subplot(3,1,3);

plot(t, x_time_comp, 'b');
plot(t, x_time_exp, 'r--');
legend("x(2t)","x(0.5t)");
xtitle("Time Scaling");
```



Experiment 2.4

```
t = -1:0.001:1;

x = exp(-t.^2).*sin(6*%pi*t);

t0 = 0.3;

x_shift = interp1(t, x, t - t0, 'linear');

x_rev = interp1(t, x, -t, 'linear');

clf;

plot(t, x, 'k');

plot(t, x_shift, 'b--');

plot(t, x_rev, 'r:');

legend("x(t)","x(t-0.3)","x(-t)");

xtitle("Time Shift and Reversal");
```